

Statistical Arbitrage

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Problem 1: The OU Process

1.

$$\begin{aligned}dX_t &= \theta(\mu - X_t)dt + \sigma dW_t, \quad \theta, \sigma > 0 \\d(e^{\theta t} X_t) &= e^{\theta t} dX_t + \theta e^{\theta t} X_t dt \\d(e^{\theta t} X_t) &= e^{\theta t} (\theta(\mu - X_t)dt + \sigma dW_t) + \theta e^{\theta t} X_t dt \\d(e^{\theta t} X_t) &= \theta e^{\theta t} \mu dt + \sigma e^{\theta t} dW_t \\e^{\theta t} X_t - X_0 &= \mu(e^{\theta t} - 1) + \sigma \int_0^t e^{\theta s} dW_s \\X_t &= \mu - (X_0 - \mu)e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dW_s\end{aligned}$$

2. We begin by investigating the stochastic term in the previous result:

$$\begin{aligned}X_t &= \alpha + \beta, \quad \alpha = \mu - (X_0 - \mu)e^{-\theta t}, \quad \beta = \sigma \int_0^t e^{-\theta(t-s)} dW_s \\ \beta &\sim N(0, \sigma^2 \int_0^t e^{-2\theta(t-s)} ds) \\ &\sim N\left(0, \frac{\sigma^2}{2\theta} [1 - e^{-2\theta t}]\right)\end{aligned}$$

$$\begin{aligned}E(X_t) &= E\left(\mu - (X_0 - \mu)e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dW_s\right) \\ &= \mu - (X_0 - \mu)e^{-\theta t} + E\left(\sigma \int_0^t e^{-\theta(t-s)} dW_s\right) \\ &= \mu - (X_0 - \mu)e^{-\theta t}\end{aligned}$$

$$\begin{aligned}\text{Var}(X_t) &= E(X_t^2) - E(X_t)^2 \\ &= E(\alpha^2 + 2\alpha\beta + \beta^2) - \alpha^2 \\ &= E(\beta^2) \\ &= \frac{\sigma^2}{2\theta} [1 - e^{-2\theta t}]\end{aligned}$$

This implies that the stationary distribution for $\lim_{t \rightarrow \infty} X_t$ is

$$\lim_{t \rightarrow \infty} X_t \sim N\left(\mu, \frac{\sigma^2}{2\theta}\right)$$

3. **Half-life of a deviation** First we define a deviation as a perturbation from an initial condition. Since this is a stochastic timeseries, we can inject this deviation at any point in the timeseries. We define its half-life λ as how long it takes for that deviation to decrease to half of its original value.

$$\begin{aligned} X_t|X_0 + d &= \mu + (X_0 + d - \mu)e^{\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dW_s \\ X_t|X_0 + d &= (X_t|X_0) + de^{-\theta t} \end{aligned}$$

We find that the deviation follows an exponential decay curve given by $e^{-\theta t}$. The half life of this is easy to measure, we have

$$\begin{aligned} \frac{1}{2} &= e^{-\theta \lambda} \\ \lambda &= \frac{\ln(2)}{\theta} \end{aligned}$$

4. **Exact Discretization:** We find the exact discretization by choosing $t = \Delta t$ in the original solution to the SDE. We find

$$\begin{aligned} X_{n+1}|X_n &= \mu + (X_n - \mu)e^{-\Delta t \theta} + \int_0^{\Delta t} e^{-\theta(\Delta t-s)} dW_s \\ &= \mu(1 - e^{-\Delta t \theta}) + X_n e^{-\Delta t \theta} + \int_0^{\Delta t} e^{-\theta(\Delta t-s)} dW_s \\ &= a + bX_n + \epsilon_k \end{aligned}$$

Here we have shown that $a = \mu(1 - b)$, $b = e^{-\Delta t \theta}$ and $\epsilon_k = \int_0^{\Delta t} e^{-\theta(\Delta t-s)} dW_s$. We have already shown (using Δt to replace t) that $\epsilon_k \sim N\left(0, \frac{\sigma^2}{2\theta}(1 - b^2)\right)$. We invert these equations to find,

$$\begin{aligned} \theta &= -\frac{\ln(b)}{\Delta t} \\ \mu &= \frac{a}{1 - b} \\ \sigma &= -\frac{2\text{Var}(\epsilon_k) \ln(b)}{\Delta t(1 - b^2)} \end{aligned}$$

Problem 2: Simulation and Estimation

1. Simulating Euler-Maruyama
2. **Maximum-Likelihood Estimation** Derive MLE of (θ, μ, σ) . Show it is equivalent to the least squares regression of X_{k+1} on X_k .

We begin by identifying the probability/likelihood of X_{n+1} given X_n which is gaussian. In order to simplify the equation, we denote $\mu_* = E(X_{n+1}|X_n)$ and $\sigma_* = \text{Var}(X_{n+1}|X_n)$.

$$P_n = P(X_{n+1}|X_n; \theta, \mu, \sigma) = \frac{1}{\sigma_* \sqrt{2\pi}} \exp\left(-\frac{(X_{n+1} - \mu_*)^2}{2\sigma_*^2}\right)$$

Thus we can assemble the (log) likelihood function L by taking the product of these possibilities for X_i

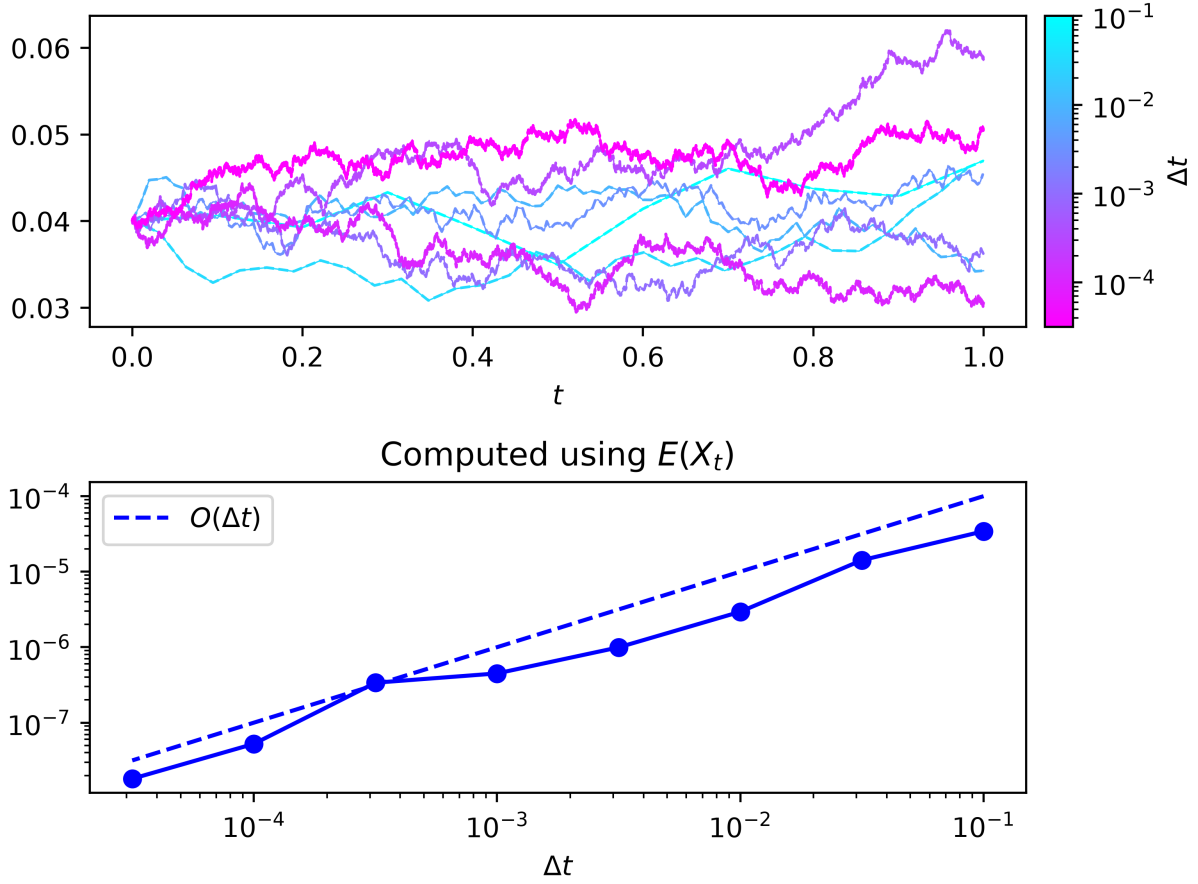


Figure 1: Several realizations of $X_t|X_0$ with varying dt

for all $i \in [0, N_t]$. We have,

$$L(\mathbf{X}_{n+1}|\mathbf{X}_n; \theta, \mu, \sigma) = \prod_{i=0}^{N_t-1} P_i$$

$$\ln(L) = \sum_{i=0}^{N_t-1} \ln(P_i)$$

$$\ln(L) = -N_t \ln(\sigma_* \sqrt{2\pi}) - \sum_i \frac{(X_{n+1} - \mu_*)^2}{2\sigma_*^2}$$

Without loss of generality, we will study the maximum likelihood of $X_1|X_0$ as the inclusion of additional terms will only add rows to the eventual linear regression problem.

$$\mathcal{L} = \ln(L(X_1|X_0; \theta, \mu, \sigma)) = -\ln(\sigma_* \sqrt{2\pi}) - \frac{(X_1 - \mu_*)^2}{2\sigma_*^2}$$

$$\mathcal{L}_{\mu_*} = \frac{(X_1 - \mu_*)}{\sigma_*^2} = 0$$

$$\mathcal{L}_{\sigma_*} = -\frac{1}{\sigma_*} + \frac{(X_1 - \mu_*)^2}{\sigma_*^3} = 0$$

From these two partial derivatives of the log-likelihood function we can solve for θ, μ, σ . From the

partial $\mathcal{L}_{\mu_*}$, we find

$$\begin{aligned}\frac{(X_1 - \mu_*)}{\sigma_*^2} &= 0 \\ X_1 - \mu - (X_0 - \mu)e^{-\Delta t \theta} &= 0 \\ X_1 &= X_0 e^{-\Delta t \theta} - \mu(1 - e^{\Delta t \theta})\end{aligned}$$

This reduces the maximum-likelihood estimation for θ, μ to a linear regression from X_0 to X_1 , i.e. we obtain (a, b) from the linear update function. The slope and y-intercept of that regression yield θ and μ .

Next we look at the equation for $\mathcal{L}_{\sigma_*}$:

$$\begin{aligned}\frac{(X_1 - \mu_*)^2}{\sigma_*^3} &= \frac{1}{\sigma_*} \\ (X_1 - X_0 e^{-\Delta t \theta} - \mu(1 - e^{\Delta t \theta}))^2 &= \sigma_*^2\end{aligned}$$

From this equation, we find that square of the residual for this linear fit is exactly equal to the “modified” variance for this update probability. To generalize this for N_t timesteps, we turn the linear regression into a least squares regression problem where we fit a single a and b (c.f. problem 1.4) to fit each timestep update. Most modern Least-Squares solvers return the sum of the squared residuals as an output alongside the linear regression. In order to compute the “modified” variance ($\sim \text{Var}(\epsilon_k)$), we divide the the sum of squared residuals by $N_t - 1$. We now have $(a, b, \text{Var}(\epsilon_k))$, and can invert the equations to solve for (θ, μ, σ) .

3. Analyze the effectiveness of θ .

Problem 3: Pair Selection and Spread

Problem 4: Time Varying Hedge Ratio

1. **Kalman-Filter** We begin by writing the state-space model,

$$\ln(P_A) = \alpha_t + \beta_t \ln(P_B) + v_t$$

where α_t and β_t are described by random walks and v_t is a zero-mean gaussian with variance R describing some noise in the price data. We will now look at the evolution of (α_t, β_t) . We know that increments to these are i.i.d. white noise. Thus,

$$\begin{bmatrix} \alpha_t \\ \beta_t \end{bmatrix} = \begin{bmatrix} \alpha_{t-1} \\ \beta_{t-1} \end{bmatrix} + d\mathbf{W}$$

Thus we expect noise to be injected into the state model with variance Δt . Let us assume that we have already done the task of fitting (α_t, β_t) for some $t \leq t_0$ with confidence P . We would then have the following prediction for t_1 :

$$\hat{\alpha}_t = (\hat{\alpha}_t, \hat{\beta}_t) = (\alpha_{t-1}, \beta_{t-1}) \hat{\mathbf{P}}_t = \mathbf{P}_{t-1} + Q$$

We then look at how closely these α_t, β_t fit the observation model: $\ln(P_A) = \mathbf{H}_t \cdot \alpha_t$. We obtain the error in our observations given by:

$$e_t = \ln(P_A) - \mathbf{H}_t \cdot \hat{\alpha}_t$$

This is coupled with the uncertainty/confidence of our our state model being injected into our observations. We find the confidence at this level being,

$$oe_t = \mathbf{H}_t \cdot (\hat{\mathbf{P}}_t \mathbf{H}_t) + R$$

and finally the ratio between our state and observation confidence being equivalent to the kalman gain K_t .

$$K_t = \frac{1}{\sigma_{e_t}} \hat{P}_t H_t$$

Using the Kalman gain we can correct our model, we have

$$\begin{aligned} \alpha_t &= \hat{\alpha}_t + K_t e_t \\ P_t &= (I - K_t H_t^T) \hat{P}_t \end{aligned}$$

2. Comparing static and Kalman hedge ratios

Problem 5: Trading Rule and Backtest

1. Trade the Standardized Set
2. Run the trading scheme on the test data
3. Diagnose and analyze a broken pair